

4. Logistic Regression

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4.1 The Model

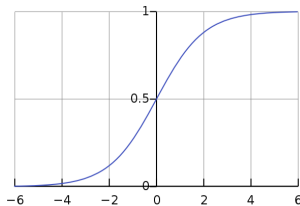
- A random variable Y is *Bernoulli distributed with parameter $p \in [0, 1]$ ($\text{Ber}(p)$)* if

$$\begin{cases} \mathbb{P}[Y = 1] = p \\ \mathbb{P}[Y = 0] = 1 - p \end{cases}$$

- Model** Let $Y \mid X \sim \text{Ber}(p(X))$, where

- $X = (X_1, \dots, X_d)$,
- $p(X) = \psi(\beta_0 + \beta_1 X_1 + \dots + \beta_d X_d)$ and
- $\psi(x) = \frac{e^x}{e^x + 1} = \frac{1}{1 + e^{-x}}$

(logistic function)



- That is,
$$\begin{cases} \mathbb{P}[Y = 1 \mid X] = p(X) \\ \mathbb{P}[Y = 0 \mid X] = 1 - p(X) \end{cases}$$
- used in binary classification; *e.g. email filtering, medical diagnoses, fraud detection, credit analytics, churn modelling, ...*

- The *probability mass function* of a $\text{Ber}(p)$ random variable Y is given by

$$f_p(y) = \begin{cases} p & \text{if } y = 1 \\ 1 - p & \text{if } y = 0 \end{cases} = p^y(1 - p)^{1-y}, \quad y \in \{0, 1\}$$

- The *probability mass function* of m independent $\text{Ber}(p)$ random variables $Y_1, \dots, Y_m$ is

$$\prod_{i=1}^m f_p(y_i) = \prod_{i=1}^m p^{y_i}(1 - p)^{1-y_i}, \quad y_i \in \{0, 1\}$$

- The *likelihood function* of $Y_1, \dots, Y_m$ is $L(p \mid y_1, \dots, y_m) = \prod_{i=1}^m f_p(y_i) = \prod_{i=1}^m p^{y_i}(1 - p)^{1-y_i}$
- The *log-likelihood function* of $Y_1, \dots, Y_m$ is

$$l(p \mid y_1, \dots, y_m) = \log L(p \mid y_1, \dots, y_m) = \sum_{i=1}^m \{y_i \log(p) + (1 - y_i) \log(1 - p)\}$$

4.2 Training/Estimation

- *Training Data:* let $(x_{i1}, \dots, x_{id}, y_i)$, $i = 1, \dots, m$, be iid realizations of $(X_1, \dots, X_d, Y)$
- Set $x_{i0} = 1$, $x_i^T b = \sum_{j=0}^d x_{ij} b_j$, $p_i(b) = \psi(x_i^T b)$ for $b \in \mathbb{R}^{d+1}$ and $i = 1, \dots, m$
- *Empirical loss minimization problem (conditional negative likelihood minimization)*

$$\begin{aligned} & \min_{b \in \mathbb{R}^{d+1}} \sum_{i=1}^m \{-y_i \log(p_i(b)) + (y_i - 1) \log(1 - p_i(b))\} \quad (\text{cross-entropy loss} \geq 0) \\ &= \min_{b \in \mathbb{R}^{d+1}} \sum_{i=1}^m \{-y_i \log(\psi(x_i^T b)) + (y_i - 1) \log(1 - \psi(x_i^T b))\} \\ &= \min_{b \in \mathbb{R}^{d+1}} \sum_{i=1}^m \left\{ y_i \log \left(\frac{1 - \psi(x_i^T b)}{\psi(x_i^T b)} \right) - \log(1 - \psi(x_i^T b)) \right\} \\ &= \min_{b \in \mathbb{R}^{d+1}} \sum_{i=1}^m \left\{ \log(1 + e^{x_i^T b}) - y_i x_i^T b \right\} \end{aligned}$$

- This is a convex minimization problem in $b \in \mathbb{R}^{d+1}$ (**Exercise**)
- Can be solved with (S)GD

4.3 Predicted Probabilities and Binary Classification

- Solution of the empirical loss minimization problem: $\hat{\beta} = (\hat{\beta}_0, \dots, \hat{\beta}_d)$
- New data point: $x = (x_1, \dots, x_d)$
- Predicted probability that $Y = 1$ conditioned on $X = (x_1, \dots, x_d)$: $\hat{p}(x) = \psi \left(\hat{\beta}_0 + \sum_{j=1}^d x_j \hat{\beta}_j \right)$
- Binary Classification
 - Choose a decision threshold $c \in (0, 1)$
 - Predict Y to be $\hat{y}(x) = \begin{cases} 1 & \text{if } \hat{p}(x) \geq c \\ 0 & \text{if } \hat{p}(x) < c \end{cases}$

4.4 Diagnostics

- **Test Data:** let $(x_{i1}, \dots, x_{id}, y_i)$, $i = m + 1, \dots, m + n$, be iid realizations of $(X_1, \dots, X_d, Y)$ independent of $(x_{i1}, \dots, x_{id}, y_i)$, $i = 1, \dots, m$
- A case i is ...
 - P (positive) if $y_i = 1$
 - N (negative) if $y_i = 0$
 - TP (true positive) if $\hat{y}_i = 1$ and $y_i = 1$
 - FP (false positive) if $\hat{y}_i = 1$ and $y_i = 0$
 - TN (true negative) if $\hat{y}_i = 0$ and $y_i = 0$
 - FN (false negative) if $\hat{y}_i = 0$ and $y_i = 1$
- **Accuracy** = $(TP + TN)/(P + N)$ *% of correctly predicted cases*
- **Imbalanced data:** often, one of the two classes P or N is much smaller than the other
 - e.g. in fraud detection, $P/(P+N)$ can be around 0.001
 - in such cases, accuracy might not be the best performance measure
e.g., just predicting N has accuracy 99.9 %

4.5 Other Performance Measures

- $\text{TPR} = \text{TP}/P$ (*true positive rate, recall, sensitivity*)

- $\text{FPR} = \text{FP}/N$ (*false positive rate*)

- ROC (*receiving operator curve*)

shows (TPR, FPR) for $c \in [0, 1]$

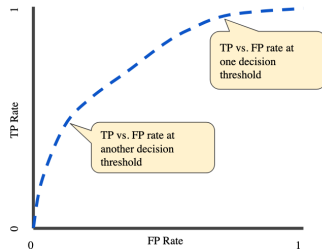
(*originally from radar detection*)

- AUC (*area under the curve*)

(*the larger the better*)

(*random guessing produces the diagonal, $\text{AUC} = 1/2$*)

(*$\text{AUC} < 1/2$ is worse than random guessing*)



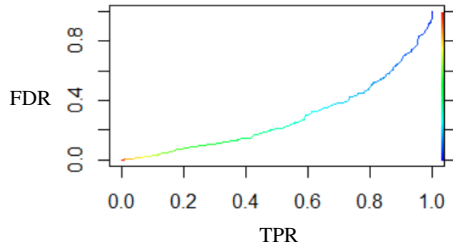
- $\text{PPV} = \text{TP}/(\text{TP} + \text{FP})$ (*positive predication value, precision*)

- $$\text{F1} = 2 \times \frac{\text{TPR} \times \text{PPV}}{\text{TPR} + \text{PPV}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \quad (\text{F1 score})$$

4.6 Credit Analytics

- Let P be a *good borrower* and N a *bad borrower*
- Try to minimize $FDR = FP/(FP + TP)$ (*false discovery rate*) *FP's result in losses*
- Try to maximize $TPR = TP/P$ (*true positive rate, recall*) *increases business volume*
- Try to obtain a flat FDR/TPR-curve

shows (FDR, TPR) for $c \in [0, 1]$



- Here, a small AUC (area under the curve) is desirable